

Nombre:

Día

Mes

Año

Folio

Tema:

Tarea 15

1- Calcular los siguientes límites:

a) $\lim_{x \rightarrow 3} (-2x^2 + 8x - 4)$

$$(-2(3)^2 + 8(3) - 4)$$

$$-18 + 24 - 4$$

$$\frac{2}{1}$$

b) $\lim_{x \rightarrow -1} \frac{2x^2 - x - 3}{x + 1}$

$$\frac{2(-1)^2 - (-1) - 3}{-1 + 1}$$

$$\frac{2 + 1 - 3}{-1 + 1}$$

$$\frac{0}{0}$$

$$\frac{0}{0}$$

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c)

$$\lim_{x \rightarrow -2} \frac{x^3 + 8}{x + 2}$$

$$= \frac{(-2)^3 + 8}{(-2) + 2}$$

$$= \frac{-8 + 8}{-2 + 2}$$

$$= \frac{0}{0}$$

Otra forma (algebraicamente)

$$\lim_{x \rightarrow -2} \frac{(x+2)(x^2-2x+4)}{x+2}$$

$$= \lim_{x \rightarrow -2} x^2 - 2x + 4$$

$$= (-2)^2 - 2(-2) + 4$$

$$= 4 + 4 + 4$$

$$= 12$$

$$\begin{array}{r} 1 \ 0 \ 0 \ 8 \\ -2 \quad \quad \quad -2 \ 4 \ -8 \\ \hline 1 \ -2 \ 4 \ 0 \end{array}$$

$$d) \lim_{x \rightarrow -1} \frac{\sqrt{2+x} - 1}{x+1}$$

$$\frac{\sqrt{2+(-1)} - 1}{-1+1} = \frac{\sqrt{1} - 1}{-1+1} = \frac{1-1}{-1+1} = \frac{0}{0}$$

$$\frac{\sqrt{2+x}+1}{\sqrt{2+x}+1} = \frac{(\sqrt{2+x})^2 - 1^2}{(x+1)(\sqrt{2+x}+1)} = \frac{2+x-1}{(x+1)(\sqrt{2+x}+1)}$$

$$= \frac{2+x-1}{(x+1)(\sqrt{2+x}+1)} = \frac{x+1}{(x+1)(\sqrt{2+x}+1)} = \frac{1}{\sqrt{2+x}+1}$$

$$\lim_{x \rightarrow -1} \frac{1}{\sqrt{2+x}+1} = \frac{1}{\sqrt{2+(-1)}+1} = \frac{1}{2}$$

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$$e) \lim_{x \rightarrow 4} \frac{x-4}{3(\sqrt{x}-2)} = \frac{4-4}{3(\sqrt{4}-2)} = \frac{4-4}{3(2-2)} = \frac{0}{0}$$

$$\frac{x-4}{3(\sqrt{x}-2)} = \frac{x-4}{3(\sqrt{x}-2)} \cdot \frac{\sqrt{x}+2}{\sqrt{x}+2} = \frac{(x-4)(\sqrt{x}+2)}{3(\sqrt{x^2}-2^2)} = \frac{(x-4)(\sqrt{x}+2)}{3(x-4)}$$
$$= \frac{\sqrt{x}+2}{3} = \frac{\sqrt{4}+2}{3} = \frac{2+2}{3} = \frac{4}{3}$$